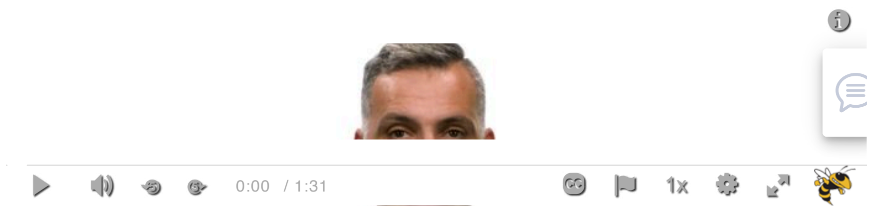
Lesson Fourteen Overview



Throughout the lecture, we are giving you some **"food for thought"** -- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will:

- Understand what "representation learning" means in the context of networks
- Study different methods to compute node and link embeddings
- Apply deep learning approaches in the context of graphs (graph neural networks) -- and explore an application (side-effects of polypharmacy)
- Gain a high-level understanding of state-of-the-art research areas in network science, including temporal networks and interdependent networks

Required and Recommended Reading

Required Reading

- **"Representation learning on graphs: Methods and applications" (https://arxiv.org/abs/1709.05584)**, by W.L.Hamilton et al. 2017

Recommended Reading

- **"Deep learning on graphs: A survey" (https://ieeexplore.ieee.org/abstract/document/9039675)**, by Z.Zhang et al. 2020
- **"Temporal networks" (https://www.sciencedirect.com/science/article/pii/S0370157312000841)**, by P.Holme and Z.Saramaki, 2012
- **"The structure and dynamic of multilayer networks" (https://www.sciencedirect.com/science/article/pii/S0370157314002105)**, by S.Boccaletti et al., 2014

Assignments

L14: Knowledge Check (https://gatech.instructure.com/courses/552990/modules/items/6023244)

L14: Quiz